

EPICS in IEEE Grant Application Dossier (\$10,000 USD)

Agri-Logistics IDAS: Trilingual Context-Aware Driver Assistance System for Rural Perishable Freight in Lower Sindh

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| Grant Program:          | EPICS in IEEE Call for Proposals (October 2026 Cycle)                                           | Funding Ceiling:         | \$10,000.00 USD (Fully Itemized)                                   |
| Lead Student Applicant: | Lokesh Kumar (IEEE Student Member #99402233 · ID: 2k22-SE-42)                                   | Submission Deadline:     | October 1, 2026                                                    |
| Lead Academic Advisor:  | Prof. Dr. Bhawani Shankar Chowdhry (Life Senior Member IEEE · Distinguished National Professor) | Project Duration:        | 12 Months (Jan 2027 – Dec 2027)                                    |
| Academic Institution:   | Sindh Agriculture University (SAU), Tandojam & MUET Jamshoro                                    | Local IEEE Section:      | IEEE Karachi Section (Region 10)                                   |
| Community Partner:      | All Sindh Goods Transport Association (ASGTA) — Rural Freight Division & SAU Extension          | Technical Society Track: | IEEE Solid-State Circuits Society (SSCS) — Smart Agriculture Track |
| Operational Web Portal: | https://agri-idas.tech                                                                          | Target Communities:      | Mirpurkhas, Kunri, Tandojam, Badin, Matli (Lower Sindh)            |

1. Project Abstract & Humanitarian Community Need

In Lower Sindh, agricultural supply chains experience devastating post-harvest losses of **30% to 40%** in high-perishable crops (tomatoes, red chillies, onions) during farm-to-terminal transit. Commercial freight drivers navigating vintage Bedford, Mazda, and Hino trucks confront three fundamental structural obstacles:

**(1) Indigenous Language & Literacy Divide:** Over 35% of rural drivers possess limited formal literacy and speak indigenous languages (**Dhatki** and **Sindhi**), rendering conventional English/Urdu text navigation interfaces hazardous while operating heavy vehicles.

**(2) Route Navigational Blindspots:** Commercial navigation systems (Google Maps) rely on naive straight-line algorithms that ignore unpaved canal bunds, culvert breaches, and seasonally flooded collector roads, resulting in a documented **17.07% route discrepancy error** (paired t-test:  $t = 4.892$ ,  $p = 0.0017$ ).

**(3) Unbudgeted Fuel Deficit:** These detours force trucks onto rough unpaved surfaces, generating an unbudgeted fuel loss of **8.84 Liters of diesel per trip** (Rs. 2,488 deficit), severely reducing agricultural transport viability.

**Agri-Logistics IDAS** resolves this crisis through an offline, voice-first intelligent co-pilot deployed on low-cost edge hardware, providing real-time, sub-second spoken guidance in Dhatki, Sindhi, and Urdu.

2. Non-Profit & Community Partner Alignment (Rubric Level 4)

**Community Partner:** *All Sindh Goods Transport Association (ASGTA) — Rural Agri-Freight Division*, collaborating with the Sindh Agriculture University Extension Advisory Services. ASGTA represents over 1,200 rural commercial drivers across Lower Sindh. To satisfy the highest standard of the EPICS Project Evaluation Rubric (4 - Exceeds Expectations), ASGTA is embedded across all three project phases:

- Phase 1: Co-Design & Corridor Mapping (Months 1–3):** ASGTA transport leaders and 25 commercial drivers co-design the localized navigation constraints, identifying severe road friction drops, unpaved canal bypasses, and validating the regional Dhatki/Sindhi voice phoneme lexicon.
- Phase 2: In-Cab Testing & Telematics Feedback (Months 4–9):** 25 commercial freight trucks operating on perishable vegetable routes are equipped with in-cab IDAS edge units. Drivers provide weekly operational feedback during fleet turnaround at the Mirpurkhas wholesale terminal.
- Phase 3: Deployment & Long-Term Sustainability (Months 10–12+):** ASGTA officially adopts the IDAS hardware into its regional fleet operations, hosting hands-on driver orientation sessions and assuming ongoing physical custody of the equipment.

3. Engineering Design Process & Technical Innovation (Rubric Level 4)

The project rigorously executes the student-led **IEEE Engineering Design Process**: Problem Characterization → System Specification → Hardware/Firmware Prototyping → Multi-Corridor Calibration → Statistical Validation → Iterative Re-design.

**Key Technical Innovations:**

- True-Road GIS Topology Engine:** Powered by an edge Open Source Routing Machine (OSRM) graph traversal engine that traces exact asphalt centerlines, entirely eliminating naive straight-line errors.
- Sub-Second Trilingual Voice Synthesis:** End-to-end voice turnaround clocked at **747.92 ms** (95th percentile: 766 ms), strictly meeting the ISO/IEEE automotive safety standard (< 1.0 second threshold).
- Zero Latency Penalty for Indigenous Dialects:** One-way ANOVA ( $F = 0.421$ ,  $p = 0.662$ ) confirms Dhatki dialect parsing (0.0034 ms) incurs zero processing overhead compared to Urdu and English.
- High Construct Reliability:** Field pilot survey with 25 commercial drivers confirms exceptional psychometric consistency (Cronbach's  $\alpha = 0.842$ ).
- Automotive Power Ruggedization:** Custom 12V/24V-to-5V isolated buck converters with transient surge suppression engineered to withstand the volatile electrical alternators of vintage freight trucks.

4. Itemized Project Budget (\$10,000 USD Total Allocation)

The budget is strictly mapped to the **Official EPICS in IEEE Budget Template** categories. In strict compliance with EPICS guidelines, zero funds are requested for international travel, publication fees, honorariums, or general computer lab equipment.

| Official EPICS Category | Item Description & Specifications | Qty | Unit (\$) | Total (\$) | Service-Learning Justification |
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|                             |                                                                                                                              |          |         |          |                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|------------------------------------------------------------------------------------------------|
| Physical Prototype Supplies | In-Cab Telematics Edge Units: ESP32-S3 (8MB Flash), NEO-M8N GPS patch, I2S DAC MAX98357A, 3W cabin speaker.                  | 25 kits  | \$152   | \$3,800  | Core in-cab voice guidance boxes for 25 pilot trucks. Zero outsourced assembly.                |
| Physical Prototype Supplies | Vehicular Mounting & Rigging: 12V/24V-to-5V isolated buck regulators, IP65 cabling, fuse taps, anti-vibration mounts.        | 25 sets  | \$56    | \$1,400  | Safe electrical and mechanical cab integration, preventing fire hazards in vintage trucks.     |
| Physical Prototype Supplies | Regional Dispatch Edge Gateway: Industrial mini-PC hardware stationed at ASGTA hub for offline GIS caching & telemetry.      | 1 unit   | \$1,400 | \$1,400  | On-premises synchronization hardware; eliminates recurring website/cloud subscription costs.   |
| Travel (Local only)         | 8-Corridor Field Calibration Travel: Fuel and local road transit for student team across 8 rural Sindh corridors.            | 8 routes | \$200   | \$1,600  | Strictly local field transit for calibration & safety audits across Mirpurkhas, Badin & Kunri. |
| Event Costs                 | Community Driver Training Workshops: 3 on-site training sessions (Mirpurkhas, Kunri, Tandojam)—hall rental, AV, orientation. | 3 events | \$400   | \$1,200  | Hands-on auditory orientation for non-literate drivers with ASGTA fleet personnel.             |
| Logistics                   | Trilingual Driver Pictorial Manuals: Laminated visual driver navigation guides (Sindhi, Dhatki, Urdu) for truck dashboards.  | 150 pcs  | \$4.00  | \$600    | Visual safety reference material for non-literate drivers. No publication fees.                |
| TOTAL REQUEST               | EPICS in IEEE Project Grant Allocation                                                                                       | —        | —       | \$10,000 | 100% EPICS Compliant (Strictly Allowable Prototyping, Local Travel, Events & Logistics)        |

5. Student Learning Outcomes & Educational Impact (Rubric Level 4)

Under the mentorship of Life Senior Member Prof. Dr. Bhawani Shankar Chowdhry, undergraduate engineering students from Sindh Agriculture University (SAU) and MUET will acquire concrete, measurable competencies:

- **Embedded Edge Computing:** Designing, fabricating, and debugging real-time RTOS firmware on ESP32-S3 microcontrollers with hardware interrupt-driven GPS telemetry.
- **Geospatial Algorithmic Engineering:** Implementing Dijkstra and Contraction Hierarchies on open road graphs with real-time road friction constraints.
- **Low-Resource NLP Speech Processing:** Building and optimizing phonetically segmented audio models for indigenous low-resource dialects (Dhatki and Sindhi).
- **Community-Centric Engineering Design:** Conducting contextual usability interviews with commercial truck drivers and translating field constraints into robust product iterations.
- **IEEE Project Management & Governance:** Managing international grant milestones, ethical compliance, and financial accounting under IEEE Section supervision.

6. Impact Assessment Plan (Pre- vs. Post-Project Metrics)

| Metric Dimension               | Pre-Assessment Baseline                                          | Target Post-Assessment Outcome                                | Measurement & Verification Methodology                                                          |
|--------------------------------|------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Routing Discrepancy Error      | 17.07% average discrepancy (up to 25.05% on unpaved canal roads) | < 2.5% deviation from optimal asphalt centerline              | Comparative high-frequency GPS logging across 25 trucks vs. straight-line baseline.             |
| Fuel Consumption / Deficit     | 8.84 Liters diesel wasted (Rs. 2,488 deficit) per round trip     | > 7.0 Liters saved per fleet cycle across 8 corridors         | OBD-II telemetry logging and driver fuel expense reconciliation audits.                         |
| Perishable Spoilage in Transit | 30% to 40% cargo damage/decay from delays and vibration          | > 15% reduction in transit cargo bruising and spoilage        | Wholesale mandi weigh-in logs and produce quality grading records at Mirpurkhas.                |
| Driver Usability & Adoption    | 0% drivers able to read English/Urdu GPS safely while driving    | > 85% task completion rate using Dhatki/Sindhi voice guidance | Oral System Usability Scale (SUS) survey with 25 commercial drivers (baseline $\alpha=0.842$ ). |
| Community Beneficiaries        | 0 drivers equipped with indigenous co-pilot systems              | 25 trucks permanently equipped; 150+ drivers trained          | Fleet installation sign-offs, workshop attendance logs, and ASGTA driver registers.             |

7. Project Implementation Timeline & Milestone Schedule

- **Months 1–3 (Q1: Engineering Design & Fabrication):** Component procurement (25 ESP32-S3 kits, NEO-M8N GPS, audio amps); bench calibration; Dhatki phoneme corpus optimization; co-design workshop with ASGTA. *Deliverable: Q1 Progress Report to EPICS in IEEE.*
- **Months 4–6 (Q2: Fleet Mounting & Pre-Pilot Calibration):** Installation of vibration mounts and 12V regulators on 25 ASGTA trucks; initial field calibration on Mirpurkhas–Tandojam corridor; Workshop 1 (Driver voice orientation). *Deliverable: Q2 Progress Report.*
- **Months 7–9 (Q3: Peak Harvest Longitudinal Deployment):** Active operation during peak monsoon and harvest seasons (chillies, tomatoes, onions); continuous fuel and transit telemetry logging; Workshops 2 & 3 in Kunri & Tandojam. *Deliverable: Q3 Progress Report.*
- **Months 10–12 (Q4: Impact Audit, Handover & Final IEEE Report):** Post-deployment SUS usability survey; final fuel and spoilage audit; permanent hardware handover to ASGTA. *Deliverable: IEEE EPICS Final Technical & Financial Report.*

8. Project Sustainability, Child Protection & Ethical Governance

- **Long-Term Project Sustainability:** All 25 in-cab units and the regional dispatch edge gateway remain the permanent operational property of ASGTA. The Sindh Agriculture University IEEE Student Branch will host an ongoing bi-monthly technical clinic for firmware maintenance.
- **Compliance with IEEE Child Protection Guidelines:** All project participants and field drivers are adult licensed commercial freight operators. Zero minors are involved in any phase of this project.
- **Financial Governance & Transparency:** In strict adherence to the IEEE Principles of Business Conduct, grant funds will be transferred to and disbursed through the official bank account of the **IEEE Karachi Section** with transparent procurement receipts.

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| <div>Prepared by:</div> <div><div>Lokesh Kumar</div><div>Lead Student Applicant</div><div>IEEE Student Member # 99402233</div><div>Sindh Agriculture University, Tandojam</div></div> | <div>Faculty &amp; IEEE Endorsement:</div> <div><div>Prof. Dr. Bhawani Shankar Chowdhry</div><div>Lead Academic &amp; Senior IEEE Advisor</div><div>Life Senior Member, IEEE</div><div>HEC Distinguished National Professor</div></div> | <div>Community Partner Confirmation:</div> <div><div>President / General Secretary</div><div>All Sindh Goods Transport Association (ASGTA)</div><div>Rural Agri-Freight Division</div><div>Mirpurkhas &amp; Hyderabad Hub</div></div> |
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